when we move relative to one another, these clocks do not agree. They fall out of synchronization; they measure different amounts of elapsed time between two chosen events. The same is true of distance. Each of us carries our own yardstick, our own monitor of distance in space. Each yardstick is equally precise, yet when we move relative to one another, these yardsticks do not agree; they measure different distances between the locations of two specified events. If space and time did not behave this way, the speed of light would not be constant and would depend on the observer's state of motion. But it *is* constant; space and time *do* behave this way. Space and time adjust themselves in an exactly compensating manner so that observations of light's speed yield the same result, regardless of the observer's velocity.

Getting the quantitative details of precisely how the measurements of space and time differ is more involved, but requires only high school algebra. It is not the depth of mathematics that makes Einstein's special relativity challenging. It is the degree to which the ideas are foreign and apparently inconsistent with our everyday experiences. But once Einstein had the key insight—the realization that he needed to break with the more than two-hundred-year-old Newtonian perspective on space and time—it was not hard to fill in the details. He was able to show precisely how one person's measurements of distances and durations must differ from those of another in order to ensure that each measures an identical value for the speed of light.[4]

To get a fuller sense of what Einstein found, imagine that Bart, with heavy heart, has carried out the mandatory retrofitting of his skateboard, which now has a maximum speed of 65 miles per hour. If he heads due north at top speed—reading, whistling, yawning, and occasionally glancing at the road—and then merges onto a highway pointing in a northeasterly direction, his speed in the northward direction will be *less* than 65 miles per hour. The reason is clear. Initially, all his speed was devoted to northward motion, but when he shifted direction some of that speed was diverted into eastward motion, leaving a little less for heading north. This extremely simple idea actually allows us to capture the core insight of special relativity. Here's how:

We are used to the fact that objects can move through space, but there is another kind of motion that is equally important: objects also move